

Amir Hossein Osooli

Education

- 2024–Present **M. Sc.**, *Biomedical Engineering*, University of Tehran, GPA: 17.68/20
Thesis: Investigation of Thermomechanical Response and Deformation Dynamics of Superficial, Muscular, and Adipose Soft Tissues under Photothermal Stimulation
Supervisor: Dr. M. H. Salimi
- 2018–2022 **B. Sc.**, *Mechanical Engineering*, University of Tehran, GPA: 15.99/20
Thesis: Applications of Blockchain Technology in Robotics
Advisor: Dr. K. Ghaemi Osgouie
- 2014–2018 **High School**, *Beheshti High School*, Mathematics and Physics, National Organization for Development of Exceptional Talents (NODET)
- 2011–2014 **Middle School**, *Beheshti Middle School*, National Organization for Development of Exceptional Talents (NODET)

Skills

- **Programming:** Python, MATLAB, C/C++, git, $\LaTeX$
- **Robotics:** Robot Operating System (ROS)/ROS 2
- **Embedded Systems:** Raspberry Pi, Arduino, ESP32
- **Web developing:** HTML
- **Mechanical Design:** Solidworks
- **Engineering:** Adams, Automation Studio, Proteus
- **Image/Video Editing:** Adobe Illustrator
- **Art:** Playing the Setar in Semi-Advanced Level

Experience

Research

- May 2025 **Graduate Research Assistant**, *Prof. M. H. Salimi*, Department of Biomedical Engineering, University of Tehran, Tehran, Iran
Present **Developing an Optical based AI enhanced Imaging system**
Tasks: *Investigating and developing the Relevant Generative Adversarial Network (GAN) Image generating Methods; Presenting the Results in a Weekly Basis*
- Jan 2025 **M. Sc. Seminar**, *Prof. M. H. Salimi, Prof. B. Vahidi*, Department of Biomedical Engineering, University of Tehran, Tehran, Iran
Aug 2025 **Advanced optical imaging methods in histology with a focus on cancer tissues, heart, and skin**
Tasks: *Doing research on relevant methods, Writing a report, Providing a defence presentation*

Dec 2023 **Research Assistant**, Prof. M. Mehrandezh, Dr. K. Ghaemi Osgouie, Department
Apr 2024 of Engineering and Applied Science, University of Regina, SK, CA

Developing a torque controller for a rope climbing robot

Tasks: Investigating the Relevant Control Methods; Presenting the Results in a Weekly Basis

Teaching Assistant (TA)

Feb 2022 **Engineering Mathematics**, Dr. V. Soltani Mohammadi, Mechanical Engineering

Jul 2022 Department, University of Tehran, Tehran, Iran

Feb 2022 **Calculus II**, Dr. V. Soltani Mohammadi, Mechanical Engineering Department,

Jul 2022 University of Tehran, Tehran, Iran

Oct 2021 **Calculus I**, Dr. V. Soltani Mohammadi, Mechanical Engineering Department,

Jan 2022 University of Tehran, Tehran, Iran

Sep 2013 **Teaching "How to Solve Rubik's cube"**, Beheshti High School, Kashan, Iran

May 2014

Internship

Aug 2021 **Mechanical Engineering Intern**, PANTOhealth, Berlin, Germany

Sep 2021 Tasks: Data Analysis and Signal Processing on Pantograph Data in MATLAB, Link to Github

Publications

Journals Articles

- Osooli, A. H, Vahidi, B, Salimi, M. H, **Dual-Wavelength photothermal speckle modulation for mapping localized absorption in turbid media.** (Under review)

Conference Proceeding

- Osooli, A. H, Salimi, M. H, (2025) **Super-Resolution Generative Adversarial Network for Photothermal Optical Coherence Tomography Signal Enhancement.** In 2025 10th International Conference of Biomedical Engineering (ICBME). Tabriz (oral presentation), Link to Presentation video, IEEE: Link to article

Projects

Fall 2025 **Robotic Rehabilitation**, Path Pulse - Smart Blind Assistance System, Link to Github

C++ (Arduino IDE 1.8.19)

Spring 2025 **Computational Fluid Dynamics 1**, Bioprinting CFD Analysis Tool, Link to Github

Python

Jan 2025 **Maze path-planning**, Link to Github

Python

July 2023 **Code Foundation for ROS**, The Construct Academy, Link to Github

Ubuntu 20.04, ROS Noetic

July 2023 **Python for Robotics**, The Construct Academy, Programming the TurtleBot to Navigate in a Maze, Link to Github

ROS Noetic

Spring 2022 **Computational Fluid Dynamics**, *Solving the one-dimensional Burgers' equation using the Keller Box Scheme*, [Link to Github](#)
MATLAB R2022a

Spring 2022 **Fundamentals of Electronics (ARDUINO)**, *Simulation of the Temperature Valve Control System with Arduino board*, [Link to Github](#)
C++ (Arduino IDE 1.8.19), Proteus 8.13

Online Courses

- **ChatGPT Prompt Engineering for Developers**, Coursera, Andrew NG
- **Python for Robotics**, The Construct, Alberto Ezquerro
- **Linux for Robotics**, The Construct, Alberto Ezquerro
- **ROS 2 Learning Week 2023**, The Construct, Alberto Ezquerro

Certifications

- **AI Agents**, Hugging Face, Joffrey Thomas, Feb 2025, [Link to the Certificate](#)
- **Writing in the Sciences**, Coursera, Kristin Sainani, Stanford University, Jan 2025, [Link to the Certificate](#)
- **Machine Learning Specialization**, Coursera, Andrew NG, Feb 2024, [Link to the Certificate](#)
- **Unsupervised Learning, Recommenders, Reinforcement Learning**, Coursera, Andrew NG, Feb 2024, [Link to the Certificate](#)
- **Advanced Learning Algorithms**, Coursera, Andrew NG, Jan 2024, [Link to the Certificate](#)
- **Code Foundation for ROS**, The Construct, Alberto Ezquerro, July 2023, [Link to the Certificate](#)
- **Supervised Machine Learning: Regression and Classification**, Coursera, Andrew NG, May 2023, [Link to the Certificate](#)
- **The Complete Python Programming Course: Beginner to Advanced**, Udemy, Joseph Delgadillo, Nick Germaine, March 2023, [Link to the Certificate](#)
- **Holo Screen Idea Presentation**, 14th Annual Training Quality Assurance System (ATQS, *NETKA* in Persian), 2015

Awards, Honors

- **Ranked 69th among approximately 71,400 participants**, in the National Graduate University Entrance Examination for Mechanical Engineering, Iran, 2024
- **Mitacs Accelerate Research Grant**, Government of Canada and Province of Saskatchewan, Amount: \$270,000 CAD, 2024
Named as research intern on competitively-awarded project
project: Smart Grain Sentinel: A Deep-Grain Inspector Robot for Precision Post-Harvest Monitoring and Loss Prevention in Agricultural Storage.
- **The Ira A. Fulton Schools of Engineering Graduate Scholarship** from Arizona State University (ASU), Amount: \$8,000, 2024
- **The Provost Master's Scholarship** from Stevens Institute of Technology, Amount: \$15,000, 2023

- **Supporter Foundation of University of Tehran Prize**, 2019
- **Ranked 2nd** Among Mechanical Engineering Students in University of Tehran, 2018-2019
- **Admitted to NODET (National Organization for Development of Exceptional Talents) Schools** Among ~2,000 candidates in Kashan, Iran, 2011
- **Won 2 Medals** in Taekwondo Hanmadang, 2011

Workshops

- **The Next Generation of Medical Imaging Systems in Biomedical Engineering**, Dr. M. H. Salimi, 10th International Conference of Biomedical Engineering (ICBME). Tabriz, November 20, 2025
- **Experiences of a Teaching Assistant**, Dr. A. Bahrami, Dr. R. Khosravi, University of Tehran, May 11, 2022
- **Application of Artificial Intelligence in Mechanical Engineering**, Dr. M. Shariatpanahi, Dr. M. Ayati, University of Tehran, June 2, 2021